impl PrivateCandidateMeasure for MaxDivergence

Michael Shoemate

Proves soundness of the `MaxDivergence` implementation of `PrivateCandidateMeasure` in `mod.rs`.

1 Hoare Triple

Precondition

`base_map` is the privacy map of a valid `MaxDivergence` measurement.

Pseudocode

```
1  def max_divergence_validate(has_threshold: bool, repetitions):
2      if has_threshold and not repetitions.is_geometric():  #
3          raise "thresholded MaxDivergence selection requires geometric repetitions"
4      if repetitions["family"] == "Poisson":  #
5          raise "Poisson selection is not supported under MaxDivergence"
6      return ()
7
8
9  def max_divergence_new_privacy_map(base_map, has_threshold: bool, repetitions):
10     if has_threshold:
11         factor = 2.0  #
12     else:
13         factor = 2.0.inf_add(repetitions["eta"])  #
14
15     def privacy_map(d_in):
16         return base_map(d_in).inf_mul(factor)  #
17
18     return privacy_map
```

Postcondition

Theorem 1.1. The `validate` method accepts exactly the supported pure-DP combinations implemented in `mod.rs`, and `new_privacy_map` returns the privacy map stated in the proof definition of `make_select_private_candidate`.

2 Proof

The validation logic is immediate:

- line 2 rejects thresholded constructions unless the repetition family is geometric;
- line 4 rejects all Poisson constructions;
- no other combinations are rejected.

For privacy accounting, there are two accepted cases.

- In the thresholded case, line 11 sets the factor to 2. By Theorem 3.1 of [1], the thresholded geometric construction is 2ϵ-DP when the base mechanism is ϵ-DP.

- In the non-thresholded case, line 13 computes an upward-rounded factor satisfying `factor` $\geq 2 + \eta$. By Corollary 3 of [2], the negative-binomial best-of-k construction is $(2 + \eta)\epsilon$-DP.

Line 16 applies this factor to `base_map(d_in)`, so the returned privacy map is a conservative upper bound on the implemented pure-DP privacy loss.

References

[1] Jingcheng Liu and Kunal Talwar. Private selection from private candidates, 2018.

[2] Nicolas Papernot and Thomas Steinke. Hyperparameter tuning with renyi differential privacy, 2021.